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Polycyclic aromatic hydrocarbons from the co-pyrolysis of catechol and propyne

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Abstract

In order to investigate C_3 species as potential participants in aromatic-ring-growth reactions of polycyclic aromatic hydrocarbons (PAH) from solid fuels, we have performed pyrolysis experiments in an isothermal laminar-flow reactor (at temperatures, 700–1000 °C; residence time, 0.3 s) with the C_3 hydrocarbon propyne, the model fuel catechol (representative of aromatic moieties in coal and biomass), and with propyne and catechol together (in a catechol-to-propyne molar ratio of 0.938). Analysis of the 1000-°C reaction products of propyne – by high-pressure liquid chromatography with diode-array ultraviolet-visible absorbance detection – has led to the identification of 58 two- to eight-ring PAH that have never before been reported as products of propyne pyrolysis or combustion. At all temperatures except 1000 °C, however, propyne-only pyrolysis produces very low yields of PAH – a consequence of low propyne (and propadiene) conversion and high production of benzene, which remains relatively invulnerable to ring-growth reactions. Catechol-only pyrolysis, in contrast, produces fairly high yields of PAH (up to 5.6% on a %-fed-carbon basis) – a result of catechol's high production of effective C_2 and C_4 growth species as well as a rich radical pool that accelerates fuel conversion and aromatic-ring-growth reactions. When catechol and propyne are co-pyrolyzed, the radical pool provided by catechol causes a dramatic acceleration of propyne's decomposition to methyl and acetylene – shifting the small-hydrocarbon product distribution from one that is rich in C_3 species to one that is rich in C_2 species. The 1- and 2-ring aromatic hydrocarbon product distribution also shifts from one dominated by unreactive benzene to one having more substituted species, more reactive to ring-growth. The net result of these shifts is a tremendous boost in PAH production: At all temperatures >800 °C, PAH yields from the co-pyrolysis experiments are 2.5–2.8 times higher than what would result from the two fuels being pyrolyzed independently.
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Keywords: Polycyclic aromatic hydrocarbons; Pyrolysis; Catechol; Propyne; PAH

1. Introduction

C_3 hydrocarbons are widely recognized as important sources of benzene in pyrolysis [1–3] and combustion [4–6] systems and have been noted [3] for their high propensities to form soot. High-molecular-weight hydrocarbons (up to 662 amu) have been observed [1] as products of the pyrolysis of the C_3 fuel propadiene, but investigations [6–9] into the role of C_3 species as participants in aromatic-ring-growth reactions have generally been limited to the formation of aromatics of no more than two rings. Little is known about the role of C_3 hydrocarbons in ring-growth reactions that transform small polycyclic aromatic hydrocarbons (PAH) into PAH of higher and higher ring number.

To address this question, particularly in the context of PAH production from solid fuels, we

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have conducted pyrolysis experiments with the C_3 hydrocarbon propyne, the model fuel catechol (*ortho*-dihydroxybenzene), and with propyne and catechol together. Catechol has been chosen since its molecular structure is representative of aromatic moieties in coal [10] and biomass fuels [11] and since the PAH produced from catechol pyrolysis have been shown [12] to be compositionally very similar to those produced from coal volatiles pyrolyzed at the same temperature. Propyne has been chosen since it is a good source [2,4,5,13] of propargyl radical, which has been proposed [6–9] as an important growth agent in the formation of 2-ring PAH from 1-ring aromatics.

The propyne pyrolysis, catechol pyrolysis, and catechol/propyne co-pyrolysis experiments are each conducted in an isothermal laminar-flow quartz-tube reactor at temperatures of 700–1000 °C and a fixed residence time of 0.3 s. The products are analyzed by gas chromatography as well as high-pressure liquid chromatography (HPLC) with diode-array ultraviolet-visible (UV) absorbance detection, a technique ideally suited for the isomer-specific analysis of PAH. In the following, we report the identities of the PAH products identified by HPLC/UV and compare PAH product yields, as functions of temperature, for the different sets of experiments. Differences in PAH production for the co-pyrolysis experiments, in comparison to the pyrolysis experiments with the individual fuels, are interpreted with reference to the differences in the distributions of the C_1–C_5 and 1- and 2-ring aromatic hydrocarbon products. Effects on PAH of different ring number are also examined.

2. Experimental equipment and procedures

The catechol pyrolysis, propyne pyrolysis, and catechol/propyne co-pyrolysis experiments are conducted in a reactor system, described elsewhere [14], that consists of a fuel vaporizer, isothermal laminar-flow quartz-tube reactor, and product collection system. Catechol pyrolysis experiments are carried out by loading catechol particles (>99.5% pure) into a Pyrex tube fixed within the vaporizer, a constant-temperature oven held at 85 °C, for slight vaporization of the catechol. A flowing stream of ultra-high purity nitrogen picks up the vapor-phase catechol, resulting in a 0.65 mol% carbon loading in the reactor feed gas. Upon exiting the vaporizer, the vaporized catechol/nitrogen mixture enters the quartz-tube reactor, which is insulated at both ends and maintained at uniform temperature by a three-zone electrically heated furnace. The reactor is operated at temperatures of 700–1000 °C and at a fixed residence time of 0.3 s.

The propyne pyrolysis experiments are carried out exactly like the catechol pyrolysis experiments except for two differences: (1) Instead of a pure nitrogen carrier gas, the carrier gas is nitrogen containing 1155 ppm of propyne. (2) The carrier gas is passed through the by-pass line in the fuel vaporizer so that no catechol is picked up. The feed to the reactor in this case is thus nitrogen with 0.3465 mol% carbon loading from propyne.

The catechol/propyne co-pyrolysis experiments are carried out exactly like the catechol pyrolysis experiments except instead of the pure nitrogen carrier gas, the carrier gas is nitrogen containing 1155 ppm of propyne. The feed to the reactor in this case is thus nitrogen with 0.65 mol% carbon loading from catechol and 0.3465 mol% carbon loading from propyne. Therefore 65.2% of the total carbon fed to the reactor in the co-pyrolysis experiments is from catechol; 34.8% is from propyne. (The corresponding ratio of moles of catechol to moles of propyne in the feed is 0.938, a level chosen to ensure that the propyne provided in the feed is almost two orders of magnitude higher than the level produced by catechol during the experiments.)

For each of the pyrolysis experiments, the products exiting the reactor are quenched [14] to room temperature. The gas-phase products are analyzed as described elsewhere [14]. The condensed-phase products are dissolved in dichloromethane, concentrated in a Kuderna-Danish evaporator, analyzed by gas chromatography with mass spectrometry (for the 1- and 2-ring aromatics) [14] and by HPLC with diode-array UV absorbance detection (for PAH of $\geqslant 3$ rings). Details of the HPLC chromatographic separation method are given elsewhere [15]. The individual PAH products are unequivocally identified by matching each component's retention time and UV absorbance spectrum with those of the appropriate reference standard. Quantification of the identified PAH comes from extensive calibration of the HPLC/UV instrument with reference standards.

Product yields obtained from repeat experiments under the same conditions give very reproducible results. In the catechol pyrolysis experiments and catechol/propyne co-pyrolysis experiments, the masses of carbon quantified in the collected C_1–C_{10} products, PAH, oxygen-containing aromatics, and unconverted fuel account for all of the carbon fed to the reactor, so a carbon balance is closed in those experiments. We also close a carbon balance on the propyne pyrolysis experiments at temperatures <950 °C. Propyne produces some soot at 950 °C and more at 1000 °C, however; so our carbon balances are 95% and 90%, respectively, for the propyne pyrolysis experiments at those temperatures, since we observe the soot but do not measure it.

3. Results and discussion

We have conducted catechol pyrolysis, propyne pyrolysis, and catechol/propyne co-pyrolysis experiments in the isothermal laminar-flow reactor at a fixed residence time of 0.3 s and at seven temperatures within the range of 700–1000 °C.

HPLC/UV analysis of the products of propyne pyrolysis at 1000 °C yields the chromatogram of Fig. 1, which shows the structures of the 61 products unequivocally identified by matching each product component's UV spectrum with that of the respective reference standard. Fig. S1 in the Supplemental Material provides an example of such a match for naphtho[2,1-*a*]pyrene, the propyne pyrolysis product eluting at 64.8 min in Fig. 1. The products in Fig. 1 range in size up to eight aromatic rings and comprise the following compound classes, as designated by structure color: 25 benzenoid PAH, in black; 8 indene benzologues, in green; 8 fluoranthene benzologues, in dark blue; 5 cyclopenta-fused PAH, in red; 7 ethynyl-substituted species (triacetylene and 6 ethynyl-substituted aromatics), in purple; 6 methylated aromatics, in light blue; and 2 bi-aryls, in gray. The names and structures of all 61 products are presented, by class, in Table S1 in the Supplemental Material. The literature reports PAH, identified with varying degrees of specificity, in the pyrolysis products of propadiene [1], propane [16], and propylene [17]. However, outside of a shock-tube propyne pyrolysis study [2] – which reports the presence of a $C_{12}H_8$ product and a $C_{14}H_8$ product that the authors infer could be PAH – our literature search has turned up no paper that reports any of the 58 two- to eight-ring PAH of Fig. 1 as products of propyne pyrolysis or combustion. It therefore appears that all of these PAH are first-time identifications as pyrolysis products of propyne.

All of the products of propyne pyrolysis in Fig. 1 are also produced in the catechol pyrolysis and catechol/propyne co-pyrolysis experiments in our reactor, but, at a given reactor condition, catechol and catechol/propyne produce additional polycyclic aromatic compounds (particularly large PAH and oxygen-containing polycyclics) that are not produced by propyne pyrolysis alone. It happens, however, that all 91 of the PAH that we have identified as products of catechol-only pyrolysis and of catechol/propyne co-pyrolysis are also the PAH that we have identified [18] in the products of catechol/1,3-butadiene co-pyrolysis. The reader is therefore referred to our paper [18] on catechol/1,3-butadiene co-pyrolysis, for the structures and names of all of the PAH products we have identified as products of catechol pyrolysis and catechol/propyne co-pyrolysis.

The PAH from the propyne pyrolysis, catechol pyrolysis, and catechol/propyne co-pyrolysis experiments have been quantified as functions of pyrolysis temperature, and the yield/temperature plots of 36 individual higher-yield PAH products appear as Supplementary Figs. S2–S7 in the Supplemental Material. The yields of the product PAH of $\geqslant 3$ rings have been summed at each temperature, and the results are plotted, as functions of temperature, in Fig. 2a for each experiment set: catechol pyrolysis (blue squares), propyne pyrolysis (red circles), and catechol/propyne co-pyrolysis (black triangles). Because different fuels are used in the different experiment sets, the total PAH yields of Fig. 2a are expressed as "% fed carbon as carbon in PAH of $\geqslant 3$ rings." For the catechol pyrolysis experiments, the "fed carbon" is the carbon in the fed catechol; for the propyne experiments, it is the carbon in the fed propyne; for the catechol/propyne co-pyrolysis experiments, it is the carbon in the fed catechol and propyne. The fourth symbol plotted in Fig. 2a is the purple diamond, which corresponds to a calculated "weighted sum," for use in comparison to the catechol/propyne co-pyrolysis experimental data. For any given temperature, the weighted sum is calculated from:

$$\text{weighted sum} = 0.652\, y_c + 0.348\, y_p \tag{1}$$

where 0.652 and 0.348 are the fractions of the feed carbon in the co-pyrolysis experiment from catechol and propyne, respectively, and y_c and y_p are the yield values from the catechol pyrolysis experiment (blue square) and propyne pyrolysis experiment (red circle), respectively. This weighted sum represents what the catechol/propyne co-pyrolysis experiment would yield if the two fuels together simply behaved as a linear combination of the two fuels individually.

Of the four PAH yield curves in Fig. 2a, we first consider only those for catechol pyrolysis (in blue) and for propyne pyrolysis (in red). Consistent with our observation that, at a given temperature, propyne produces fewer large PAH than catechol, Fig. 2a reveals that PAH yields from propyne pyrolysis are much lower than those from catechol at all temperatures investigated. PAH yields from catechol become significant at temperatures as low as 850 °C and rise with temperature until 950 °C, above which they start to decline. PAH production from propyne, on the other hand, is extremely low at temperatures below 950 °C and climbs to an appreciable level only at our highest temperature of 1000 °C.

To account for the differences in PAH yields, between the two fuels, we turn to Figs. 2b and c. The blue curves of Fig. 2b show the temperature-dependent yields, from catechol-only pyrolysis, of unreacted catechol (the fed catechol not converted) and three main product groups: CO plus phenol, summed C_1–C_5 hydrocarbons, and summed 1- and 2-ring aromatic hydrocarbons. The red curves of Fig. 2c show the temperature-dependent yields, from propyne-only pyrolysis, of unreacted propyne (the fed propyne not converted) and two main product groups, summed C_1–C_5 hydrocarbons and summed 1- and 2-ring aromatic hydrocarbons. Comparing the blue unreacted-catechol yields in Fig. 2b with the red unreacted-propyne yields in Fig. 2c, we see that catechol conversion begins at a lower temperature than does propyne conversion

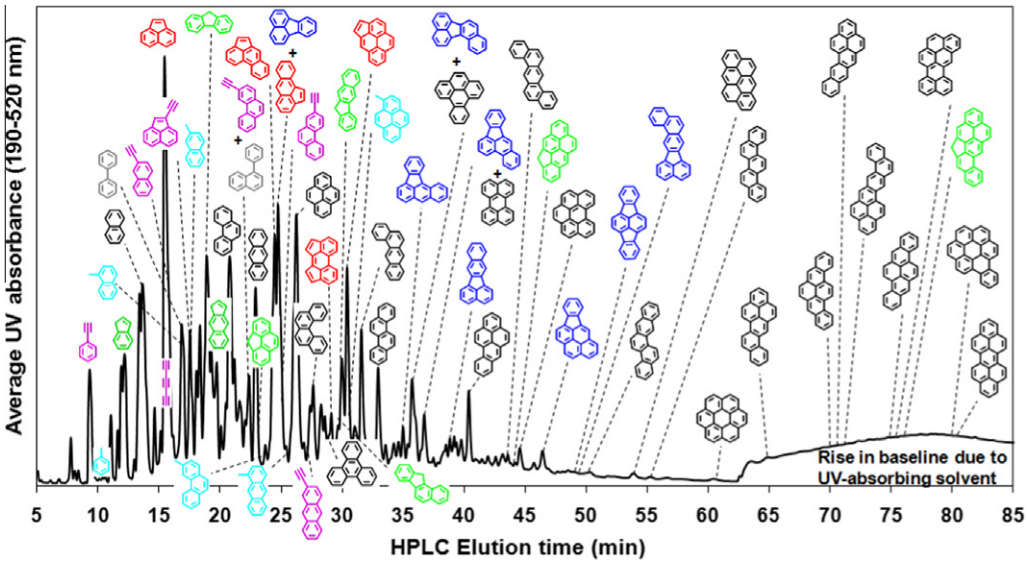

Fig. 1. HPLC chromatogram of the products of propyne pyrolysis at 1000 °C and 0.3 s. All product names and structures, color-coded by class, are presented in Table S1 in the Supplemental Material.

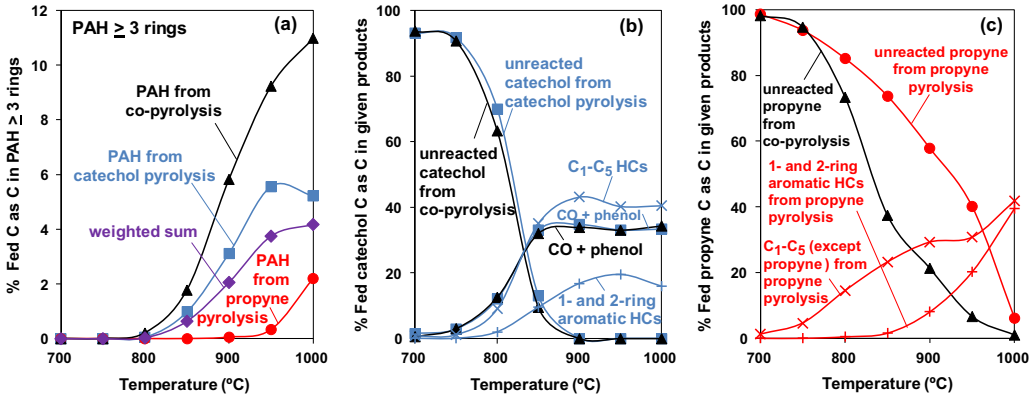

Fig. 2. Major product yields and fuel conversions, as functions of temperature, for catechol pyrolysis (blue symbols and curves), propyne pyrolysis (red symbols and curves), catechol/propyne co-pyrolysis (black symbols and curves). Weighted sums (purple symbols and curves) from Eq. (1). (a) Summed yields of PAH of ⩾3 rings; (b) catechol conversion and product-group yields; (c) propyne conversion and product-group yields.

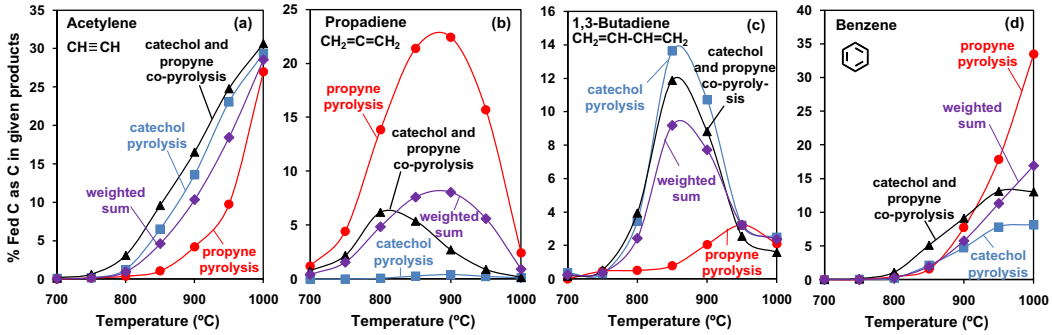

Fig. 3. Yields, as functions of temperature, of (a) acetylene; (b) propadiene; (c) 1,3-butadiene; (d) benzene. Experiments: catechol pyrolysis (blue squares); propyne pyrolysis (red circles); catechol/propyne co-pyrolysis (black triangles). Weighted sum values (purple diamonds) calculated from Eq. (1).

and that catechol conversion increases much more rapidly with temperature (in the 750–850-°C range) than does propyne conversion. Both of these observations are directly related to the hydroxylic oxygen of catechol's structure. First, the O—H bond is the weakest bond in the catechol molecule, having a bond-dissociation energy (BDE) of 81.2 kcal/mol [19], which is lower than 88.9 kcal/mol [19], the BDE for the weakest bond in propyne, the methyl C—H bond. Catechol's lower BDE enables the radical-initiation step to take place at a lower temperature for catechol than for propyne. Second, once catechol pyrolysis reactions start, OH and other vigorous oxygen-containing radicals are generated, accelerating catechol's conversion, primarily to CO and small hydrocarbons. No such oxygen is present in the propyne pyrolysis environment, so even once initiated, propyne conversion to all other product hydrocarbons proceeds relatively slowly, as the red curves of Fig. 2c demonstrate. The net result is that for any given temperature between 800 and 950 °C, much more fuel carbon is available for PAH and PAH-precursor species in the catechol pyrolysis environment than in the propyne pyrolysis environment – even though Fig. 2b shows that one-third of fed catechol carbon ends up being "tied up" as CO and phenol.

Catechol's enhanced conversion, relative to that of propyne, is not the only reason for the higher PAH yields, in Fig. 2a, from catechol pyrolysis than from propyne pyrolysis. Differences in the compositions of the C_1–C_5 hydrocarbon products and the 1- and 2-ring aromatic hydrocarbon products also play major roles. The yield/temperature plots for all of these individual hydrocarbon products of our experiments appear in the Supplemental Material as Figs. S8–S10. Fig. 3 presents such plots for four key C_2–C_6 product species: acetylene, propadiene, 1,3-butadiene, and benzene. (The color and symbol conventions explained for Fig. 2a also apply to Fig. 3.) As we have previously reported [14,18] (and is partially evident in Fig. 3), 1,3-butadiene is the single highest-yield hydrocarbon product of catechol pyrolysis at temperatures <900 °C; at temperatures >900 °C, acetylene is. Since ethylene and vinylacetylene are also major products of catechol pyrolysis [14,18], C_2 and C_4 hydrocarbons together account for ⩾84% of catechol's total C_1–C_5 hydrocarbon products, throughout the 700–1000-°C temperature range of our experiments. As others [20] have shown for other fuels – and as we have shown for catechol [18] – C_2 and C_4 species are extremely effective as growth agents for PAH. It is also the case that among the 1- and 2-ring aromatic hydrocarbon products of catechol, a substantial portion contains methyl, vinyl, or ethynyl substituents, which enhance these aromatic products' reactivity toward further growth, relative to their unsubstituted counterparts.

Unlike catechol, propyne produces only small amounts of 1,3-butadiene, as shown in Fig. 3c, and even smaller amounts of vinylacetylene (Supplementary Fig. S9 in the Supplemental Material), so C_4 growth species are not abundant in the propyne pyrolysis environment. Neither are C_2 species at temperatures ⩽900 °C; above 900 °C, the high yields of acetylene in Fig. 3a suggest that the rates of Reaction 2 [4,5,13] are appreciable.

$$\mathrm{H + H_3C{-}C{\equiv}CH \rightarrow \underset{\text{methyl}}{\dot{C}H_3} + \underset{\text{acetylene}}{HC{\equiv}CH}} \qquad (2)$$

Fig. 3b shows that propyne's main pyrolysis product at temperatures <950 °C is its isomer, propadiene, which forms either by reaction with H atom [4,13] or by unimolecular rearrangement [21]. The high yields of propadiene (in Fig. 3b) and of unreacted propyne (in Fig. 2c) cause these two C_3 species alone to account for >80% of the carbon fed to the reactor in the propyne pyrolysis experiments at all temperatures ⩽900 °C; they account for >95% at all temperatures ⩽850 °C.

At temperatures ⩾900 °C, Reaction 3 [2,4], the formation of propargyl radical from propyne, apparently becomes significant, since at these temperatures, the red curve of Fig. 3d shows a marked

$$\mathrm{H + H_3C{-}C{\equiv}CH \rightarrow H_2 + \underset{\text{propargyl}}{H_2\dot{C}{-}C{\equiv}CH}} \qquad (3)$$

increase in the yield of benzene, almost certainly the product (primarily) of propargyl/propargyl recombination [4–6]. Even though benzene is produced in high amounts (accounting, at all temperatures, for ⩾85% of the total 1- and 2-ring aromatic hydrocarbon products from propyne pyrolysis), its carbon remains virtually unavailable for PAH formation or growth, due to benzene's remarkable stability. Its invulnerability to reaction is evident in Fig. S3 in the Supplemental Material, which shows that – despite the high yields of benzene in the propyne pyrolysis experiments – bi-phenyl yields for propyne pyrolysis are virtually zero at temperatures ⩽950 °C. It is only at the highest temperature of 1000 °C, when benzene's aryl C—H bond can be broken (as evidenced by bi-phenyl formation) – and when acetylene yields are high and 1,3-butadiene yields are appreciable – that PAH yields from propyne pyrolysis also become appreciable.

We therefore see that there are a number of reasons why PAH production is higher from catechol pyrolysis than from propyne pyrolysis: (1) Catechol conversion occurs much more rapidly than propyne conversion. (2) Catechol produces high levels of C_2 and C_4 species that are effective growth agents for PAH; propyne does not, except at the highest temperature, 1000 °C. (3) Catechol produces a wide variety of 1- and 2-ring aromatic hydrocarbons, many of them with aliphatic sub-

stituents that enhance reactivity toward molecular growth; propyne's 1- and 2-ring aromatic products, on the other hand, are dominated by benzene, the least-reactive member in this product group. (4) The catechol reaction environment contains oxygen-containing radicals that increase the radical pool in general, accelerating all pyrolysis reactions, including those responsible for PAH growth. No such oxygen-containing radicals are present in the propyne-only pyrolysis environment.

Having examined the differences, in PAH yields, between the catechol pyrolysis and propyne pyrolysis experiments, we now consider PAH yields from catechol/propyne co-pyrolysis, which are plotted as the black triangles in Fig. 2a. Fig. 2a shows that for all temperatures at which PAH are formed, PAH yields are much higher for the co-pyrolysis experiments than for either propyne-only or catechol-only pyrolysis. For each temperature in the range of 850–1000 °C, the PAH yield from the catechol/propyne co-pyrolysis experiment is 2.5–2.8 times the respective weighted sum, the yield that would be produced if the two fuels together behaved as a linear combination of the two fuels individually.

To account for the huge enhancement in PAH yields observed for catechol/propyne co-pyrolysis, we again refer to Figs. 2b and c and 3. We first consider fuel conversion. Fig. 2b shows that, at each temperature, the yield of unreacted catechol – expressed as the percent of fed *catechol* carbon that exits the reactor as carbon in unreacted catechol – is the same for the catechol-only pyrolysis experiments (blue curve) as it is for the catechol/propyne co-pyrolysis experiments (black curve), so propyne's presence in the feed to the reactor in the co-pyrolysis experiments has no effect on catechol's conversion. It also has no effect on the percentage of fed catechol carbon going to CO and phenol, as the black and blue curves for the summed yields of these products are also coincident in Fig. 2b. Turning to Fig. 2c, however, we see that at temperatures >750 °C, propyne conversion is tremendously enhanced when catechol is present – i.e., the black unreacted-propyne curve is much lower than the red unreacted-propyne curve. (Catechol pyrolysis produces only a very small amount of propyne – the maximum yield occurring at 900 °C [14] and corresponding to ~1.9 on the 0–100 *y*-axis scale of Fig. 2c – so propyne production from catechol itself has no noticeable contribution to the black curve of Fig. 2c.) We see then that when catechol is present in the propyne pyrolysis environment, the rich pool of oxygen-containing (and other) radicals from catechol have the effect of greatly accelerating propyne's conversion: in the 800–900-°C temperature range, propyne conversion is almost two times higher in co-pyrolysis than when propyne is pyrolyzed alone. Furthermore, the rich radical pool from catechol appears to have the same effect on propyne's isomer, propadiene. In the same temperature range of 800–900 °C, Fig. 3b shows that propadiene yields from the co-pyrolysis experiments – unlike those from the propyne-only pyrolysis experiments – fall significantly with increasing temperature. These drastic depletions of propyne and propadiene in the co-pyrolysis experiments translate into a drastic reduction of C_3 species in the pyrolysis environment: At 850 °C, C_3 hydrocarbons compose 97% (on a carbon basis) of the total hydrocarbon species in the propyne-only pyrolysis environment; in the catechol/propyne co-pyrolysis environment, they compose 28%. At 900 °C, the values are 84% and 12%, respectively.

Because Reactions 2 and 3 are the two main channels of propyne conversion (other than isomerization) during propyne-only pyrolysis – and since both reactions involve radical reactants – we would expect the radical-pool-induced enhancement of propyne conversion in the co-pyrolysis experiments to translate into increased yields of the products of Reactions 2 and 3, especially at temperatures ⩽900 °C, where propyne concentrations are still quite high, even in the co-pyrolysis environment. Evidence that our expectation is correct comes from Fig. 3, which presents the yields, from the co-pyrolysis experiments, of some of the products of Reactions 2 and 3, along with the corresponding weighted sums, which represent the product yields that would come from the catechol/propyne co-pyrolysis experiments if catechol pyrolysis and propyne pyrolysis proceeded independently. Consistent with a radical-pool-induced enhancement of Reaction 2, Fig. 3a shows that acetylene yields from the co-pyrolysis experiments (in black) are substantially higher than the weighted sums (in purple) – more than 100% higher at temperatures ⩽850 °C and tapering off to about 30% higher at 950 °C. The corresponding enhancement of methyl production from Reaction 2 is evident from Supplementary Figs. S8, S10, and S2, which show that yields of methane, ethane, toluene, and the methylnaphthalenes from the co-pyrolysis experiments are all substantially higher than their weighted sums for temperatures between 800 and 950 °C.

Consistent with a radical-pool-induced enhancement of Reaction 3, Fig. 3d shows that at temperatures <950 °C, the co-pyrolysis yields of benzene, the product of propargyl/propargyl recombination, are substantially higher than the weighted sums. However, the difference between the co-pyrolysis benzene yield and the weighted sum decreases as temperature rises, until finally the two curves cross and the co-pyrolysis yield at 1000 °C lies below the weighted sum value for that temperature. We believe the reasons for the declining difference are two-fold. First, the rate of benzene production by propargyl/propargyl recombination is second order in propargyl concentration, but – due to the facilitated conversion

(in Fig. 2c) of propyne in the co-pyrolysis experiments (much of it to acetylene) – propargyl concentrations in the co-pyrolysis experiments at temperatures >950 °C would not be expected to be high. Second, even though benzene can form by reactions other than propargyl/propargyl recombination in the co-pyrolysis environment, the rich radical pool associated with the co-pyrolysis environment would increase the opportunity for benzene to undergo hydrogen abstraction and convert to phenyl radical. (The huge jump in bi-phenyl yields, in Supplementary Fig. S3, for the co-pyrolysis experiments provides evidence for phenyl production in the co-pyrolysis environment.) Unlike benzene, phenyl is very reactive and could participate in PAH growth reactions.

In addition to the effects on acetylene and benzene, we also make the following observations from the co-pyrolysis experiments. First, Supplementary Figs. S2 and S10 show that among the 1- and 2-ring aromatic hydrocarbons, those with methyl, vinyl, or ethynyl substituents exhibit higher enhancements in their yields, due to co-pyrolysis, than do their unsubstituted counterparts. These C_1 and C_2 substituent groups are more reactive than the aromatic structures to which they are attached, so the increase in the yields of the substituted 1- and 2-ring aromatics benefits subsequent molecular growth to PAH. Second, we do not see, among our products, any evidence of C_3-subsituted aromatics (e.g., propynylbenzene) that would indicate propargyl addition to an aromatic ring structure – and yields of indene (which theoretically could form by reaction of propargyl with phenyl [9]) in the co-pyrolysis experiments are only marginally larger than the weighted sums. We do, however, see lots of vinyl- and ethynyl-substituted 1- and 2-ring aromatics – as well as ethynyl-substituted and cyclopenta-fused PAH – all indicative of C_2 addition. These observations – along with the tremendously enhanced yields of acetylene (and ethylene and ethane) and the greatly depleted yields of propyne and propadiene in the co-pyrolysis experiments – lead us to conclude that it is primarily the C_2 species that are the growth agents responsible for the tremendous enhancement in PAH production in the catechol/propyne co-pyrolysis environment.

Consistent with successive C_2 additions being responsible for PAH growth in the catechol-propyne co-pyrolysis environment, we find, as illustrated by Fig. S11 in the Supplemental Material, that the ratio of the co-pyrolysis yield to the weighted sum increases with PAH ring number. Those ratios vary somewhat with pyrolysis temperature, but averaged over the appropriate ranges of temperature, they are: 2-ring, 1.4 ± 0.2; 3-ring, 2.3 ± 0.3; 4-ring, 4.2 ± 0.7; 5-ring, 4.8 ± 1.3; 6-ring, 5.7 ± 0.6; 7-ring, 6.9 ± 1.2. Because the synergistic effects of co-pyrolysis affect the largest PAH most, these effects could have implications for PAH growth and soot formation in a variety of fuel and combustion systems.

We therefore see that there are four main factors contributing to why PAH yields from catechol/propyne co-pyrolysis are 2.5–2.8 times higher than what the two fuels would produce independently: (1) facilitation of propyne's conversion when it is in the presence of the rich radical pool generated by catechol pyrolysis; (2) shifting of the small-hydrocarbon product distribution from one that is rich in C_3 species to one that is rich in C_2 species, effective PAH growth agents; (3) shifting of the 1- and 2-ring aromatic hydrocarbon product distribution from one dominated by unreactive benzene to one having more substituted species, which are more reactive towards subsequent growth reactions; (4) an increased ability, with the enhanced radical pool, to abstract aryl hydrogen from benzene and other unsubstituted aromatics, forming phenyl and other aryl radicals, which are more reactive toward ring-growth reactions than their respective molecular counterparts.

Finally, we acknowledge that combining the C_3-rich propyne pyrolysis environment with the C_2- and C_4-rich (but C_3-deficient) catechol pyrolysis environment opens up pathways for reactions not open when the fuels are pyrolyzed separately. For example, propargyl from propyne and vinylacetylene from catechol could react to form benzyl [1,6], which could combine with propargyl to form naphthalene [6–8]. Similarly, naphthylmethyl and propargyl radicals could combine to form phenanthrene and anthracene – and higher-ring-number arylmethyl radicals with propargyl could form still larger PAH. The introduction of reactions involving such "cross combinations," not active in either the catechol-only or propyne-only pyrolysis environments, could help explain why PAH production from catechol/propyne co-pyrolysis is even higher than from catechol pyrolysis alone.

4. Conclusion

The results of our study show that at 850–1000 °C, the temperatures at which PAH production is significant in our experiments, PAH yields from the co-pyrolysis of catechol and propyne are 2.5–2.8 times higher than what would result if the two fuels were pyrolyzed independently. On the surface, this finding appears similar to our finding [18] that over the same range of temperatures, PAH yields from the co-pyrolysis of catechol and 1,3-butadiene are 2.1–2.5 times higher than what would result if these two fuels were pyrolyzed independently. However, the means through which the catechol/1,3-butadiene and catechol/propyne co-pyrolysis systems achieve their elevated PAH yields – and how these means

affect the yields of particular PAH in the two co-pyrolysis systems – differ substantially.

In the case of catechol and 1,3-butadiene – because C_2 and C_4 species dominate the hydrocarbon products of both catechol pyrolysis and 1,3-butadiene pyrolysis – adding 1,3-butadiene to the catechol pyrolysis environment essentially just enhances the levels of the C_2 and C_4 species that catechol produces on its own. Since these added C_2 and C_4 species are in the radical-rich catechol pyrolysis environment, their roles as aromatic growth agents are realized, and there is effectively an "across-the-board" acceleration of the overall progression from small hydrocarbons to one- and two-ring aromatics to PAH. Because C_4 species are the dominant growth agent in the catechol/1,3-butadiene co-pyrolysis environment [18], PAH of all ring numbers are found to benefit equally from the synergistic effects.

In the case of catechol and propyne – because catechol pyrolysis produces very low yields of C_3 hydrocarbons [14,18] and because the main product of propyne pyrolysis at temperatures <950 °C is another C_3 species, propadiene – adding catechol to the propyne pyrolysis environment causes huge compositional changes to the reaction environment. The radical pool provided by catechol causes a dramatic acceleration of propyne's decomposition to methyl and acetylene – shifting the small-hydrocarbon product distribution from one that is rich in C_3 species to one that is rich in C_2 species. The one- and two-ring aromatic hydrocarbon product distribution also shifts from one dominated by unreactive benzene to one having more substituted species, more reactive to ring growth. The net result of these shifts is a tremendous boost in PAH production, and because C_2 species are the dominant growth agent in the catechol/propyne co-pyrolysis environment, the synergistic enhancement of PAH yields increases with increasing PAH ring number.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.proci.2010.06.072.

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